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Airbag ECU

Messung und Verarbeitung der Sensoreingänge zur Auslösung des Airbags
Measurement and Processing of Sensor Signals to Process an Airbag Operation

Phase 1: Item Definition

Version: 1.00

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Contents

1 Meetings within the project

Topic	Time	Room	Participants
Item Definition	14:00	Zoom	ALL
Bericht:	*none*		

2 Unresolved topics

No unresolved topics

3 Glossar

Tabelle 1: Glossar

Begriff / Term	Erklaerung / Explanation
ACU	Airbag Control Unit the ECU responsible for crash detection and airbag deployment control
ADC	Analog-to-Digital Converter converts analog sensor voltage to a digital integer value
ARM Cortex-M3	32-bit RISC processor core used in the Arduino Due (SAM3X8E)
ASIL	Automotive Safety Integrity Level A to D, D being highest requirement (ISO 26262)
CAN	Controller Area Network automotive serial communication bus standard (ISO 11898)
ECU	Electronic Control Unit embedded computing module in a vehicle
ESD	Electrostatic Discharge overvoltage event that can damage electronic components
EMI	Electromagnetic Interference unwanted electrical noise affecting circuit operation
GPIO	General Purpose Input/Output configurable digital pin on a microcontroller
HW	Hardware
ISO 26262	International standard for functional safety of road vehicle electrical/electronic systems
LDO	Low Drop-Out Regulator linear voltage regulator with low input-output voltage differential
MCU	Microcontroller Unit
MEMS	Micro-Electro-Mechanical System miniature sensor technology used for accelerometers and gyroscopes
MOSFET	Metal-Oxide-Semiconductor Field-Effect Transistor used as switching element in output drivers
Safing Condition	A secondary, independent confirmation of crash event required before deployment is enabled (dual-point safety architecture)
SWR	Software Requirement functional requirement assigned to the software module
SW	Software
TVS	Transient Voltage Suppressor protection diode for input overvoltage clamping

4 Item Definition

The item under analysis is the Airbag ECU the central controller responsible for keeping vehicle occupants safe during a crash.

At its core, this ECU does one critical job: decide, fast and correctly, whether to deploy restraints. It continuously monitors incoming data from both peripheral sensors (impact and pressure) and onboard g-sensors, using this redundant sensing architecture to cross-validate crash severity before triggering airbags or seatbelt pretensioners.

The goal of this analysis is threefold defining the ECU's system boundaries, conduct a Hazard Analysis and Risk Assessment (HARA), and establish ASIL-rated Safety Goals (SGs) that will directly shape the Technical Safety Concept going forward.

What makes this ECU uniquely challenging is the nature of its operating environment: it must process sensor data, run decision logic, and execute deployment commands all within milliseconds, with no margin for error. Redundancy isn't optional here; it's the foundation of the entire safety architecture.

4.1 System Objectives

Tabelle 2 System Objectives

Objective
Detect frontal, side, and rollover impacts
Deploy airbags and seatbelt pretensioners within milliseconds
Ensure high reliability and fault tolerance
Maintain operation during power interruption (backup energy)
Communicate with other vehicle ECUs

4.2 System Architectural Overview

A granular evaluation of the ECU's technical architecture is mandatory to identify the malfunction pathways that compromise safety requirements. In an ASIL D environment, we must account for every potential failure point within the hardware and communication blocks.

The Airbag ECU technical architecture integrates the following elements:

Processing Units: A high-performance microcontroller featuring a dedicated CPU, an external Watchdog for execution monitoring, and a Backup Power circuit (capacitive energy storage) to ensure deployment capability even after battery disconnection during an impact.

Communication Interfaces: The system relies on the Serial Peripheral Interface (SPI) for synchronous communication between the microcontroller (acting as the SPI Master) and its peripheral sensing/actuation ICs, alongside CAN bus integration for vehicle-level diagnostics.

Sensing Array: A multi-domain array utilizing redundant sensing paths, including upfront impact sensors, side-impact pressure sensors, and internal tri-axial acceleration sensors.

Actuation Path: High-side and low-side power stages that control the firing loops for the deployment of airbags and pretensioners.

The Serial Peripheral Interface (SPI) is a critical focus of this analysis. As the SPI Master, the ECU's internal shifter logic and baud rate generator are vital; any corruption here such as a fire command issued due to corrupted noise or a wrong decode of clock selection can lead directly to an unintended deployment. This architectural complexity necessitates a structured approach to Hazard Identification.

Figure 1: Airbag ECU System Architecture Block Diagram — showing all sensors, Satellite Sensor Interface, Sensor SPI Bus, Main CPU, Watchdog, Flash/EEPROM, CAN Transceiver, LIN Transceiver, Safing Engine, Firing SPI Bus, all deployment drivers and actuators.

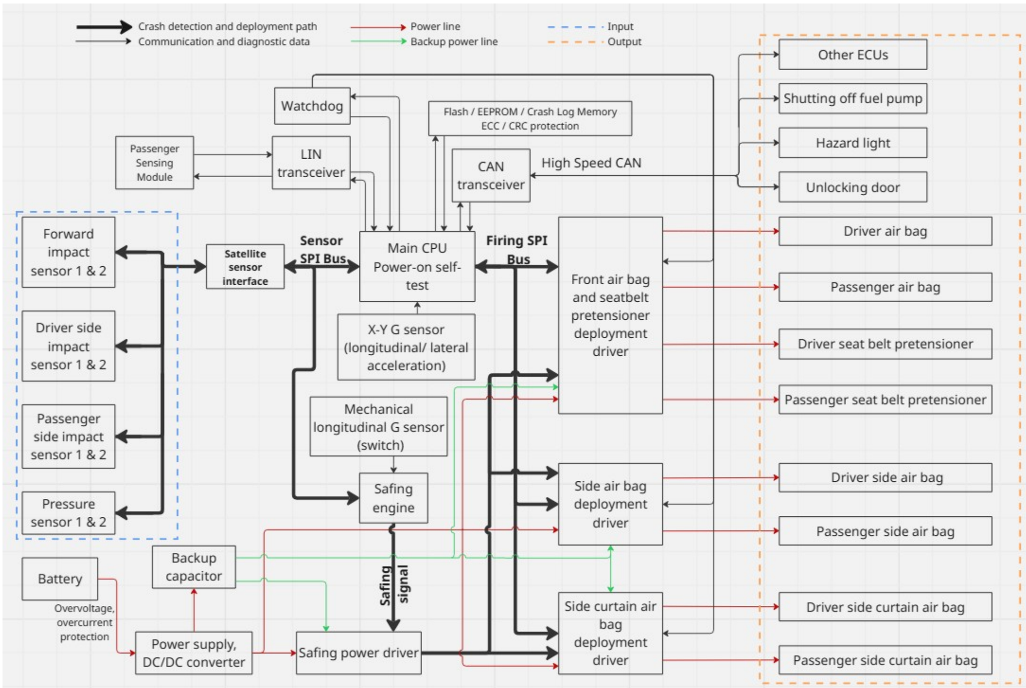

Abbildung 1 Airbag ECU System Architecture Block Diagram (Source: ECU.png, 03_System_Architecture)

Figure 2: Architecture Version 2 — Baby Seat Sensing Module and Passenger Sensing Module added to LIN transceiver path.

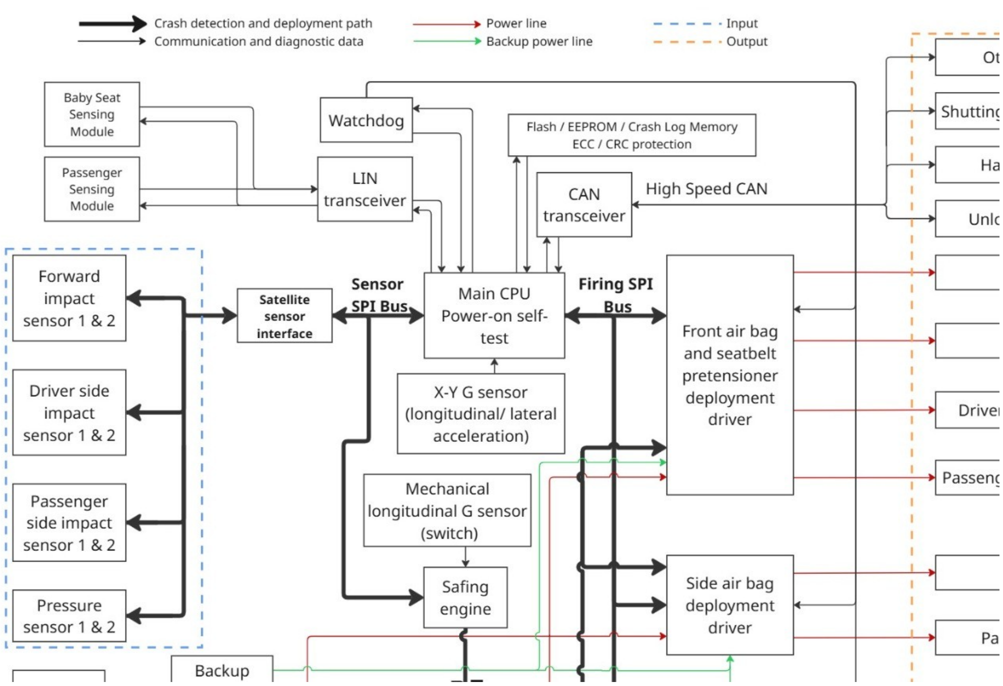

Abbildung 2 : Airbag ECU System Architecture V2 (Source: Architecture.jpeg, 03_System_Architecture)

4.3 System Subsystems

Tabelle 3 System Subsystems

Subsystem	Description
Sensor Subsystem	Forward impact sensors (1 and 2), Driver-side impact sensors (1 and 2), Passenger-side impact sensors (1 and 2), Pressure sensors (1 and 2), X-Y G-Sensors (longitudinal and lateral acceleration), Mechanical longitudinal G sensor (hardware switch backup). Sensors communicate via Sensor SPI Bus. Satellite sensors connect through a Satellite Sensor Interface.
Processing Unit (Main CPU)	Performs Power-On Self-Test (POST). Continuously evaluates sensor data. Runs crash detection algorithms. Controls deployment decisions. Connected modules: Watchdog (system monitoring), Flash/EEPROM (crash logs and configuration), CAN and LIN communication interfaces.
Communication Interfaces	CAN Transceiver (High-Speed CAN): communicates with other ECUs, sends crash status and diagnostics. LIN Transceiver: interfaces with passenger sensing module and baby seat sensing module.
Safety Subsystem	Watchdog: monitors CPU health and resets on failure. Safing Engine: independent validation of crash conditions, validates crash condition separately from the main CPU, generates the safing signal, operates in parallel with the main CPU. Mechanical G Sensor: hardware-related redundant trigger.
Deployment Subsystem	Firing SPI Bus: secure communication between CPU and deployment drivers. Deployment Drivers: Front airbag and seatbelt pretensioners driver, Side airbag deployment driver, Side curtain airbag deployment driver.
Actuators (Output)	Driver airbag, Passenger airbag, Driver seatbelt pretensioner, Passenger seatbelt pretensioner, Side airbags (driver and passenger), Curtain airbags (driver and passenger).
Power Supply System	Battery Input (primary power source, overvoltage/overcurrent protection), DC/DC converter (regulates voltage for ECU components), Backup Capacitor (provides energy during power loss e.g. crash disconnection), Safing Power Driver (controls energy delivery to deployment circuits).

4.4 Interfaces

Tabelle 4 Interfaces

Interface	Type	Purpose
Sensor SPI Bus	Input	Sensor data acquisition
Firing SPI Bus	Output	Deployment control
CAN	Bidirectional	Vehicle communication
LIN	Bidirectional	Passenger sensing module

4.5 Crash Event Data Flow

Tabelle 5 Crash Event Data Flow

Step	Action
1	Impact detected by multiple sensors
2	CPU runs crash algorithm
3	Safing engine validates event independently
4	Deployment is allowed only when both conditions are satisfied: Main CPU confirms the crash event AND Safing Engine asserts the safing signal
5	Airbags and pretensioners deploy
6	Crash data stored in Flash memory
7	CAN messages sent to other ECUs (e.g., Fuel cut-off, hazard lights, door power lock)

4.6 Safety Features

Tabelle 6 Safety Features

Safety Feature
Dual-path validation (CPU and Safing Engine)
Redundant sensors (electronic and mechanical)
Watchdog supervision
ECC/CRC memory protection
Backup energy source for deployment feasibility
Hardware-based safing signal gating

4.7 Compliance Considerations

Tabelle 7 Compliance Considerations

Standard / Requirement
ISO 26262 — Functional Safety
ASIL-D requirements for airbag system
Automotive EMC and reliability standards

5 Figure and table listing

6 List of dependent documents

No.	Dokument name	Version	Date	File name
1	Hazard Analysis and ASIL Determination for Airbag ECU	V02	02.06.2026	Hazard Analysis and ASIL Determination for Airbag ECU V02.pdf
2	Hazard Analysis and ASIL Determination for Airbag ECU	V01	07.05.2026	Hazard Analysis and ASIL Determination for Airbag ECU_V01.pdf
3	ECU Airbag System Architecture Documentation	V02	27.05.2026	ECU Airbag system architecture documentation V2.pdf
4	ECU Airbag System Architecture Documentatio	V01	07.05.2026	ECU Airbag system architecture documentation_V01.pdf
5	Airbag ECU Hardware Concept Design Report	V02	27.05.2026	AirbagECU_HW_Concept_Design_Report_V02.pdf

7 Literature

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8 Version history

Document name	date	Change log
Airbag_ECU_Phase_1	02.06.2026	v1.00 Initial version. Item Definition compiled from ECU Airbag System Architecture Documentation V2 (27.05.2026) and Hazard Analysis and ASIL Determination for Airbag ECU V02 (02.06.2026).